



SARDAR PATEL UNIVERSITY
BCA (SEMESTER-I) EXAMINATION(NC) February-2022
11 February, 2022, Friday
03:00 p.m. to 05:00 p.m.
US01FBCA02 Mathematics I

Marks: 70
[10]

Q. 1 Multiple Choice Questions

- 1 Let $A = \{6, 7, 8, 9, 10\}$ $B = \{2, 3, 6, 9, 11\}$ then $(A \cap B) =$ _____
 (a) $\{6, 9\}$ (b) $\{2, 3, 6\}$ (3) $\{6, 7, 9\}$ (d) $\{6\}$
- 2 The number of elements in the power set of a set $\{a, b, c, d\}$ are:
 (a) 0 (b) 8 (c) 16 (d) 32
- 3 Let $A = \{x: x = 2^n, n \in \mathbb{N}\} = \{2, 4, 8, \dots\}$ then A is closed under _____
 (a) multiplication (b) addition (c) both (d) none
- 4 An element e is said to be left identity element for binary operation * if
 (a) $a * e = e$ (b) $e * a = e$ (c) $a^{-1} * a = e$ (d) $a * a^{-1} = e$
- 5 Let $u = (2, 4)$, $v = (1, 2)$ then $3u - 4v =$ _____
 (a) (2, 4) (b) (3, 2) (c) (4, 8) (d) (1, 2)
- 6 If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix}$ then $A + B =$ _____
 (a) $\begin{pmatrix} 0 & 2 \\ 5 & 5 \end{pmatrix}$ (b) $\begin{pmatrix} 0 & -2 \\ -5 & 5 \end{pmatrix}$ (c) $\begin{pmatrix} 0 & 2 \\ -5 & -5 \end{pmatrix}$ (d) none
- 7 A square matrix A is said to be skew symmetric if _____
 (a) $A + A^T$ (b) $A = -A^T$ (c) $A = A^T$ (d) none
- 8 Median of 3, 7, 11, 9, 13, 1, 7, 12, 18, 6, is
 (a) 9 (b) 7 (c) 8.5 (d) none
- 9 _____ is the value dividing the observations in two equal parts
 (a) mode (b) mean (c) median (d) harmonic mean
- 10 The mean of first 10 natural numbers is
 (a) 5 (b) 5.5 (c) 6.5 (d) 1

Q. 2 Fill in the Blanks and True or False.

[08]

- 1 If $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, $B = \{4, 5\}$ then $(A/B) \cap U =$ _____
- 2 $(a^{-1})^{-1} =$ _____
- 3 If $u = (2, -3, 6)$, $v = (8, 2, -3)$ then $u \cdot v =$ _____
- 4 The observation whose frequency is highest is called _____
- 5 $D = \text{diag}(d_{ij})$ is diagonal if its non diagonal elements are all zero. (True/False)
- 6 An operation * on a set S is said to be commutative if $a * b = b * a$. (True/False)
- 7 $(B \cup C) \cap A = (B \cap A) \cup (C \cap A)$. (True/False)
- 8 Geometric mean of x, y, z is given by $\sqrt[3]{xyz}$ (True/False)

Q. 3 Answer the Following. (Any 10)

[20]

- 1 If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 3, 5\}$ $B = \{2, 4, 6\}$ and $C = \{5, 6, 7, 8\}$ then find $(A \cup B) \cap C$
- 2 State Demorgan's laws.
- 3 Let $B = \{1, 2\}$. Is B closed under (a) multiplication (b) addition (c) division?
- 4 Define * on Q by $a * b = a + b - ab$. Is it commutative?
- 5 Let $u = (1, 2, 3, 4)$, $v = (3, -4, 1, 2)$ and $w = (-4, -3, 1, 2)$ then find $4u + 3w - v$
- 6 Find x and y if $x(1, 1) + y(2, -1) = (1, 4)$
- 7 Determine those value of k for which $\begin{pmatrix} k & k \\ 4 & 2k \end{pmatrix} = 0$
- 8 Find the median for the following data:
 $X_i: 20 \ 16 \ 14 \ 13 \ 12 \ 10$
 $F_i: 7 \ 9 \ 3 \ 8 \ 5 \ 6$

- 9 The intelligence quotients(IQ's) of 10 boys is given below:
70,120, 110, 101 ,88, 83 ,95, 98, 107,100.Find mean IQ.
- 10 Define Geometric mean.
- 11 For rational number a and b Define $a*b=ab/2$. Find identity element for given binary operation *.
- 12 Write down power set of {3,4,5}

Q.4 Answer the Following.(Any 4)

[32]

- 1 Prove that
 $P(n): 1^2+2^2+\dots+n^2 = n(n+1)(2n+1)/6$
- 2 If Q is defined by $Q(a,b) = \begin{cases} 0 & \text{if } a < b \\ Q(a-b, b) & \text{if } a \geq b \end{cases}$
Then find $Q(2,3)$, $Q(19,5)$
- 3 Let $f(x) = 2x^2+4x+4$ and $g(x) = 3x^3+6x+3$ Find roots of $f(x)$ over Z_{10} and Find roots of $g(x)$ over Z_3
- 4 Let Q be the set of all rationals and * defined on Q by $a*b = a+b - ab$,
(1) Is $(Q, *)$ a simigroup?
(2) Find the identity element for *
(3) Find $3*4$, $2*(-5)$.
(4) Is $(Q, *)$ commutative?
(5) Is $(Q, *)$ monoid?
- 5 Find x if $B = \begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix}$ is symmetric
- 6 If $A = \begin{pmatrix} 2 & 6 & 2 \\ 5 & -1 & 3 \\ 3 & 1 & -2 \end{pmatrix}$ then find $A.A^T$
- 7 Calculate Mean , Median and Mode for the following data.

Weight (lbs)	130	135	140	145	146	148	149	150	157
X									
No. of persons	3	4	6	6	3	55	2	1	1

- 8 . From the following data calculate the missing value when Mean is 115.86

X	110	112	113	117	?	125	128	130
f	25	17	13	15	14	8	6	2

— X —
(2)